

Course plan for Bachelor's programme 2025/2026

Department of Advanced Computing Sciences, Maastricht University

Course code:	BCS2540
Course title:	Numerical Methods
Bachelor:	Computer Science
Course coordinator/examiner (department):	Martijn Boussé / Başak Sakçak
Teaching period:	Block 2.5
Credits in ECTS:	4

1. General provisions of course assessment

Assessment at DACS is regulated by the *Education and Examination Regulations* (EER). Some of the main points therein are:

- Every course gets a single *final grade*, which is an integer from the range [1–10].
- A student passes the course, if and only if the *final grade* is 6.0 or higher.
- A resit of a course is provided during the same academic year. The details of the extent and form of the resit can vary from the regular exam, and is provided in Table 3 of this document.
- Student is allowed to inspect the grading of every assessment component of the course. The inspection of the final grade is regulated by EER: students may, within 10 working days after the result (grade) for the course has been made official, inspect their assessment, and the required standards based on which the assessment has been done.

2. Graded components of the course assessment

During the teaching and examination period of the course, the following components will be graded, and will count towards the final grade of the course:

- **Final Written Exam (A1)** with a passing norm of 5.5 (i.e., a score of 5.5 or higher is required in order to pass the course).
- **Midterm Matlab Practical Exam (A2)** which is a (single) onsite examination moment organized during the final weeks of the course period.
- **Weekly Quizzes (A3)** which consists of multiple online graded quizzes taking place during the course period via Canvas. The grade is equal to the weighted average of the quiz grades.
- **Bonus (A4)** of 1 point can be obtained by
 - (1) achieving for 8 out of 10 tutorials the following requirements:
 - prepare the pre-tutorial task,
 - be physically present,
 - and actively participate in tutorial activities;

- (2) completing the onboarding quiz before the deadline;
- (3) completing the Matlab Onramp online course before the deadline.

If all three requirements are fulfilled, the student obtains the bonus point. (The final grade is capped at 10.)

3. Calculation of the final grade

If a student [passes A1] **and** [has the requirement for A4], then their Final Grade equals:

$$0.7 \times A1 + 0.15 \times A2 + 0.15 \times A3 + 1.$$

If a student [passes A1] **and** [does not have the requirement for A4], then their Final Grade equals:

$$0.7 \times A1 + 0.15 \times A2 + 0.15 \times A3.$$

If a student [fails A1], then their Final Grade equals:

$$1 \times A1.$$

4. Relation of the assessment to the learning outcomes of the course and to the Bachelor program

Intended learning outcomes (ILOs)	Teaching & learning activities (TLAs)	Assessment component	PQs
Knowledge and understanding: Summarise fundamental numerical algorithms of calculus, data interpolation and approximation, and linear algebra.	Lectures, tutorials and pre-tasks.	A1	1,3
Applying knowledge and understanding: Application of algorithms to simple problems from calculus, linear algebra and data science.	Tutorials and pre-tasks.	A1, A2, A3	5,6,8,9
Implementation and testing of algorithms in Matlab.	Tutorials and pre-tasks	A2	
Making judgements: Judging applicability, performance and reliability of algorithms.	Tutorials and pre-tasks	A1, A2, A3	14,16
Communication: Present solutions to problems relating to numerical computation using appropriate terminology.	Tutorials and pre-tasks	A1, A2, A3	18a
Learning Skills: Identify deficiencies in knowledge and use of programming environments, and remedy these using documentation/help facilities.	Tutorials and pre-tasks	A1, A2, A3	19,20

Table 1: Course intended learning outcomes (ILOs), corresponding teaching and learning activities (TLAs, corresponding assessment components, and the covered qualifications of the whole Bachelor program)

5. Form of the assessment components and their grading

Assessment component	Form and extent (length)	Grading scheme/key	Assessor
Final Written Exam (A1)	Written exam worth 70% of total with passing norm (5.5 or higher), closed book with formula sheet. Standard calculator permitted. 2 hours.	Detailed grading scheme.	Examiners
Midterm Matlab Practical Exam (A2)	Computer-based Matlab practical exam worth 15% of total. Students use own laptops with SchoolYear installed. 1½ hours.	Automated unit tests.	Examiners
Weekly Quizzes (A3)	Weekly online quizzes taking place during the period worth 15% of total. Students take the quizzes at home. 45 minutes per quiz.	Automated grading.	Examiners
Bonus (A4)	Bonus point (1 point). See above for requirements.	Qualitative grading based on effort put in to attempt and tutorial.	Examiners and teaching assistants

Table 2: Assessment components, form of assessment, grading scheme, and assessor

For the resit, some assessment components may be assessed differently. The details are given in Table 3.

Examination component	Form of the resit
Final Written Exam (A1)	Written Resit Exam in default resit period.
Midterm Matlab Practical Exam (A2)	Matlab Practical Resit Exam after the Period in which the course was offered. This resit is only available if the student's overall grade is a failing grade AND [their A2 grade is between 4 and 5.5 OR they have a valid excuse for not being present for the first sit of A2]. The validity of the excuse is determined by the Board of Examiners and the request should be submitted as soon as possible but no later than the start of Period 6.
Weekly Quizzes (A3)	The Weekly Quizzes are a form of asynchronous testing, i.e., the quizzes are available for several working days throughout the period and can be taken at any time (in the available time) and anywhere. If a student does not fill out the quiz before the deadline, they receive a zero for that quiz. The A3 grade is equal to the weighted average of the quiz grades. The Weekly Quizzes can be repaired only if the student's overall grade is a failing grade AND their A3 grade is between 4 and 5.5. In that case, additional repair quiz(zes) will be offered.
Bonus (A4)	There will be no opportunity to resit the bonus component.

Table 3: Resit forms of all examination elements

- ☒ Digital assessment
- ☐ TestVision
- ☐ Ans
- ☒ Ans paper exam

6. Impact of AI / IT tools

Could AI / IT tools impact quality of (some of) the assessment components of the course, and the corresponding intended learning outcomes?

☒ No

Please, explain why:

A1 and A2 represent 85% of the final score and *are invigilated*.

AI tools could potentially impact A3 as this involves weekly online quizzes that are offered asynchronously and can therefore be made at home where students can use any tools at their disposal. However, the impact is limited to 15% of the final grade, while the other 85% (A1 and A2) are invigilated ***AND A1 involves a passing norm***. Hence, the potential impact on A3 is outweighed by the benefits of continuous testing and time-on-task via retrieval practice, interleaving and spacing that is offered by a (windowed) weekly quizzing approach.

A4 is non-mandatory/optional and graded on an effort-only basis, so there is no advantage in using AI / IT tools in terms of the grade.

7. Additional provisions and remarks

☐ There are changes in the course descriptions w.r.t. the previous lecture year.

☒ There are changes in the form of examination w.r.t. the previous lecture year.

A student cannot pass the course unless the passing norm on the Final Written Examination is achieved.

The rationale for the grading of the Bonus is to encourage students to adequately prepare for the tutorials, and to have some written work as a basis for formative assessment. It is *not* intended to assess students' competence, which will be done in the examination

We have included Weekly Quizzes that will be offered via the Learning Management System and automatically graded with minimal additional feedback via the grading system. The weekly quizzes constitute 15% of the total grade. They are meant to offer multiple low-stakes assessment moments continuously throughout the course to improve time-on-task and facilitate good learning behavior (retrieval practice, spacing, interleaving, etc.).